

Equilibrium

- The reaction of nitrogen and hydrogen to form ammonia is maintained at equilibrium in a rigid container at fixed temperature:

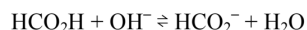
$$\text{N}_2(\text{g}) + 3 \text{H}_2(\text{g}) \rightleftharpoons 2 \text{NH}_3(\text{g})$$
Helium gas is introduced into the system, causing the total pressure to increase by a factor of two. How will the equilibrium for the reaction shift?
 - It will be unaffected.
 - It will shift to the left.
 - It will shift to the right.
 - The direction that the equilibrium will shift depends on whether the reaction is endothermic or exothermic.
- Cacodylic acid, $(\text{CH}_3)_2\text{AsO}(\text{OH})$, has a $\text{p}K_{\text{a}}$ of 6.25. At what concentration of cacodylic acid would a solution of it have a $\text{pH} = 4.00$?
 - $1.0 \times 10^{-4} \text{ M}$
 - $3.2 \times 10^{-4} \text{ M}$
 - $5.6 \times 10^{-3} \text{ M}$
 - $1.8 \times 10^{-2} \text{ M}$
- The decomposition of PCl_5 is endothermic. Which change will result in an increase in the value of K_{eq} ?

$$\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \quad \Delta H^\circ > 0$$
 - Increasing temperature
 - Increasing volume
 - I only
 - II only
 - Both I and II
 - Neither I nor II
- What concentration of carbonate ion must be achieved to precipitate 99.99% of the silver ion from a 0.100 M solution of silver nitrate? The K_{sp} of Ag_2CO_3 is 8.1×10^{-12} .
 - $8.1 \times 10^{-10} \text{ M}$
 - $8.1 \times 10^{-7} \text{ M}$
 - $1.3 \times 10^{-4} \text{ M}$
 - $8.1 \times 10^{-2} \text{ M}$
- What is the pH after 0.100 mol of NaOH is added to 250.0 mL of 0.300 M H_3PO_4 ? For H_3PO_4 , $\text{p}K_{\text{a}1} = 2.12$, $\text{p}K_{\text{a}2} = 7.21$, and $\text{p}K_{\text{a}3} = 12.38$.
 - 1.82
 - 2.24
 - 6.73
 - 6.91
- Zinc hydroxide is a sparingly soluble compound ($K_{\text{sp}} = 4.5 \times 10^{-17}$) that dissolves in strongly basic solution due to the formation of the $\text{Zn}(\text{OH})_4^{2-}$ complex ion ($K_{\text{f}} = 5.0 \times 10^{14}$). What is the minimum concentration of sodium hydroxide needed in 1.0 L of a solution which could be added to 0.10 mol $\text{Zn}(\text{OH})_2$ to fully dissolve it?
 - 2.1 M
 - 2.3 M
 - 4.2 M
 - 4.6 M
- Which statements must be correct at equilibrium?
 - The rate constants for the forward and reverse reactions are equal.
 - The rates of the forward and reverse reactions are equal.
 - I only
 - II only
 - Both I and II
 - Neither I nor II
- What is the equilibrium constant expression for the reaction below?

$$\text{AgCl}(\text{s}) + 2 \text{NH}_3(\text{aq}) \rightleftharpoons \text{Ag}(\text{NH}_3)_2^+(\text{aq}) + \text{Cl}^-(\text{aq})$$
 - $K_{\text{eq}} = \frac{[\text{AgCl}][\text{NH}_3]}{[\text{Ag}(\text{NH}_3)_2^+][\text{Cl}^-]}$
 - $K_{\text{eq}} = \frac{[\text{AgCl}][\text{NH}_3]^2}{[\text{Ag}(\text{NH}_3)_2^+][\text{Cl}^-]}$
 - $K_{\text{eq}} = \frac{[\text{Ag}(\text{NH}_3)_2^+][\text{Cl}^-]}{[\text{AgCl}][\text{NH}_3]^2}$
 - $K_{\text{eq}} = \frac{[\text{Ag}(\text{NH}_3)_2^+][\text{Cl}^-]}{[\text{NH}_3]^2}$
- Phosphoric acid has $\text{p}K_{\text{a}}$ values of $\text{p}K_{\text{a}1} = 2.15$, $\text{p}K_{\text{a}2} = 7.20$, and $\text{p}K_{\text{a}3} = 12.35$. Which species is most abundant at $\text{pH} = 11.5$?
 - H_3PO_4
 - H_2PO_4^-
 - HPO_4^{2-}
 - PO_4^{3-}
- A buffer is made by mixing 0.900 mol $\text{Na}[\text{CH}_3\text{CO}_2]$ and 0.500 mol HCl in enough water to make 2.50 L of solution. What is the pH of this solution? The $\text{p}K_{\text{a}}$ of CH_3COOH is 4.75.
 - 4.49
 - 4.65
 - 4.85
 - 5.01
- A sealed container with a volume of 1.00 L contains an equilibrium mixture of $\text{N}_2\text{O}_4(\text{g})$ and $\text{NO}_2(\text{g})$ at a given temperature. Which quantities will increase if the volume of the container is increased to 2.00 L while maintaining the same temperature?
 - The number of moles of $\text{NO}_2(\text{g})$
 - The partial pressure of $\text{NO}_2(\text{g})$
 - I only
 - II only
 - Both I and II
 - Neither I nor II
- A solution is 0.05 M in each of the anions fluoride (F^-), carbonate (CO_3^{2-}), and phosphate (PO_4^{3-}). As calcium ion is added slowly to this solution, in what order will the solids precipitate? The K_{sp} values are: CaF_2 , 3.9×10^{-11} ; CaCO_3 , 3.8×10^{-9} ; $\text{Ca}_3(\text{PO}_4)_2$, 1.0×10^{-25} .
 - First CaF_2 , then CaCO_3 , then $\text{Ca}_3(\text{PO}_4)_2$
 - First CaF_2 , then $\text{Ca}_3(\text{PO}_4)_2$, then CaCO_3
 - First $\text{Ca}_3(\text{PO}_4)_2$, then CaF_2 , then CaCO_3
 - First $\text{Ca}_3(\text{PO}_4)_2$, then CaCO_3 , then CaF_2

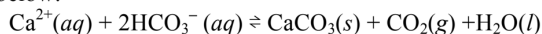
Equilibrium

13. The K_a of formic acid is 1.8×10^{-4} . What is the equilibrium constant for the reaction below?



- (A) 5.6×10^{-11} (B) 5.6×10^3
(C) 1.8×10^{10} (D) 5.6×10^{17}

14. Solid calcium carbonate is in equilibrium as described below:



If $\text{CO}_2(\text{g})$ is added to this system in a rigid container, which changes will take place?

- I. Additional $\text{CaCO}_3(\text{s})$ will precipitate.
II. The pH will increase.

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

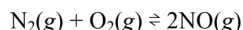
15. Calcium carbonate occurs in two common crystalline forms, calcite ($K_{sp} = 3.4 \times 10^{-9}$) and aragonite ($K_{sp} = 6.0 \times 10^{-9}$). Which of these is most thermodynamically stable?

- (A) Calcite
(B) Aragonite
(C) The two forms are equally stable.
(D) The relative stability cannot be determined from the K_{sp} values.

16. What is the pH of a 0.043 M solution of HI?

- (A) 0.04 (B) 1.37
(C) 1.67 (D) 3.15

17. At 1200 K, the equilibrium constant for the reaction



is $K_p = 2.9 \times 10^{-7}$. A mixture of 0.79 bar N_2 and 0.21 bar O_2 is prepared at 300 K and heated to 1200 K while maintaining a constant total pressure of 1.00 bar. What is the equilibrium partial pressure of $\text{NO}(\text{g})$ at 1200 K?

- (A) 2.4×10^{-8} bar (B) 1.1×10^{-4} bar
(C) 2.2×10^{-4} bar (D) 5.4×10^{-4} bar

18. Sodium sulfate is slowly added to 1.00 L of a solution that is 0.10 M in both $\text{Pb}^{2+}(\text{aq})$ and $\text{Sr}^{2+}(\text{aq})$. How much Na_2SO_4 must be added to induce precipitation of SrSO_4 ? The K_{sp} of PbSO_4 is 1.8×10^{-8} and the K_{sp} of SrSO_4 is 3.5×10^{-7} .

- (A) 3.3×10^{-6} mol (B) 3.5×10^{-6} mol
(C) 5.9×10^{-4} mol (D) 9.5×10^{-2} mol

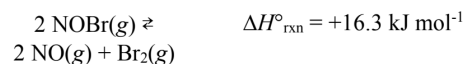
19. Lead(II) fluoride is sparingly soluble ($K_{sp} = 4.0 \times 10^{-8}$). Which, when added to a saturated solution of PbF_2 , would increase its solubility to the greatest extent?

- (A) KNO_3 (B) $\text{Pb}(\text{NO}_3)_2$
(C) KF (D) HNO_3

20. The pH of a 0.045 M solution of the monoprotic acid boric acid is 5.30. What is the K_a of boric acid?

- (A) 1.1×10^{-4} (B) 5.0×10^{-6}
(C) 2.0×10^{-9} (D) 5.6×10^{-10}

21. NOBr decomposes to NO and Br_2 gas in a closed container as shown.

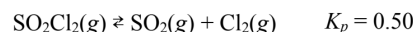


Which changes will result in an increase in the number of moles of NO present at equilibrium?

- I. Increasing the temperature
II. Decreasing the volume of the container

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

22. At a certain temperature, SO_2Cl_2 decomposes into SO_2 and Cl_2 as shown below.



An equal number of moles of SO_2Cl_2 , SO_2 , and Ar are introduced into a sealed flask to give a total pressure of 1.80 bar. The flask is maintained at the same temperature until equilibrium is attained. What is the final pressure of $\text{Cl}_2(\text{g})$?

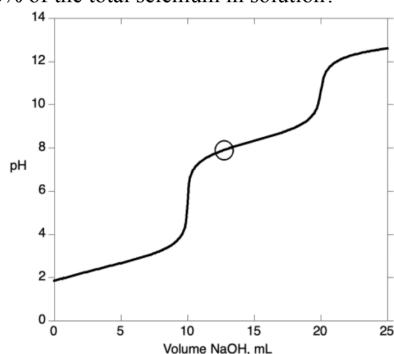
- (A) 0.23 bar (B) 0.27 bar
(C) 0.35 bar (D) 0.50 bar

23. A buffer solution consists of 0.100 mol Na_3PO_4 and 0.100 mol Na_2HPO_4 dissolved in water. How much of which component of the buffer must be added to change the pH to 12.25? The $\text{p}K_a$ of HPO_4^{2-} is 12.37.

- (A) 0.032 mol Na_2HPO_4 (B) 0.032 mol Na_3PO_4
(C) 0.132 mol Na_2HPO_4 (D) 0.132 mol Na_3PO_4

Equilibrium

24. A 0.100 M aqueous solution of H_2SeO_3 is titrated with 1.000 M NaOH solution. At the point marked with a circle on the titration curve, which species comprise at least 10% of the total selenium in solution?



- (A) H_2SeO_3 only (B) H_2SeO_3 and HSeO_3^-
 (C) HSeO_3^- only (D) HSeO_3^- and SeO_3^{2-}
25. What mass of lead(II) fluoride ($K_{\text{sp}} = 4.0 \times 10^{-8}$) will dissolve in 1.00 L of water?
 (A) 0.025 g (B) 0.049 g
 (C) 0.087 g (D) 0.53 g
26. The K_{a} of the pyridinium ion, $\text{C}_5\text{H}_5\text{NH}^+$, is 5.9×10^{-6} . What is the pH of a 0.170 M solution of pyridine, $\text{C}_5\text{H}_5\text{N}$?
 (A) 3.00 (B) 4.77 (C) 8.77 (D) 9.23
27. Excess solid silver chloride is added to water and the mixture stirred until equilibrium is achieved. Addition of which substance will increase the concentration of chloride ion in this solution?
 (A) AgNO_3 (B) AgCl
 (C) NH_3 (D) H_2O

Problems 28 and 29 are about the exothermic reaction of iodine with iodide ion shown below.



28. 1.0×10^{-4} mol I_2 and 4.0×10^{-3} mol KI are dissolved in water at 25°C to give 100.0 mL of solution. What is the equilibrium concentration of I_2 in this solution?

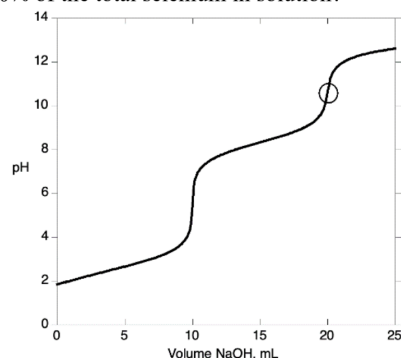
- (A) 1.3×10^{-6} M (B) 3.4×10^{-5} M
 (C) 5.0×10^{-4} M (D) 5.3×10^{-3} M

29. Which changes will result in an increase in the number of moles of $\text{I}_2(\text{aq})$ present at equilibrium?

- I. Increasing the temperature
 II. Replacing the KI with an equal mass of NaI

- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

30. A 0.100 M aqueous solution of H_2SeO_3 is titrated with 1.000 M NaOH solution. At the point marked with a circle on the titration curve, which species comprise at least 10% of the total selenium in solution?



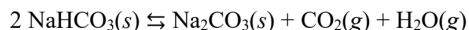
- (A) H_2SeO_3 only (B) H_2SeO_3 and HSeO_3^-
 (C) HSeO_3^- and SeO_3^{2-} (D) SeO_3^{2-} only

31. Lead(II) iodide has a solubility product constant of 1.4×10^{-8} . What is the concentration of I^- in a saturated solution of PbI_2 ?

- (A) 1.2×10^{-4} M (B) 2.4×10^{-4} M
 (C) 1.5×10^{-3} M (D) 3.0×10^{-3} M

Equilibrium

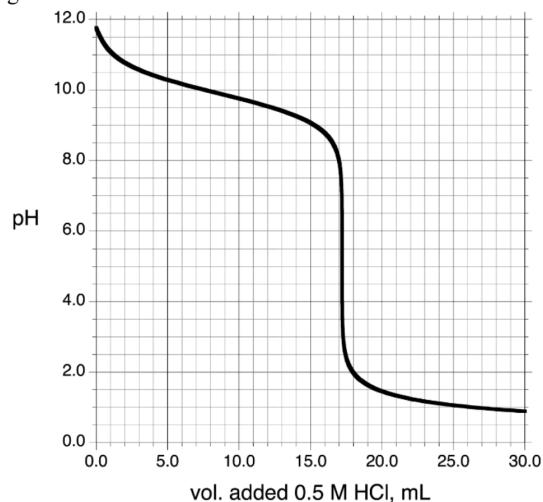
32. Sodium bicarbonate decomposes at high temperatures according to the following equation:



At a certain temperature, 50.0 g of NaHCO_3 is allowed to react in a 3.0 L evacuated vessel. At equilibrium, the total pressure is 6.25 atm, with some of both solids still present. If 100.0 g of NaHCO_3 is allowed to react in the same vessel at the same temperature, what will the pressure be at equilibrium?

- (A) 6.25 atm (B) 8.84 atm
(C) 12.5 atm (D) 25.0 atm
33. In a solution prepared by dissolving 0.1 mol NaNO_2 in 1.0 L pure water, which species has the lowest concentration?
- (A) Na^+ (B) NO_2^- (C) HNO_2 (D) H_3O^+
34. A saturated solution of which silver salt has the highest concentration of Ag^+ ?
- (A) AgCl , $K_{\text{sp}} = 1.8 \times 10^{-10}$
(B) Ag_2CrO_4 , $K_{\text{sp}} = 1.1 \times 10^{-12}$
(C) AgBr , $K_{\text{sp}} = 5.0 \times 10^{-13}$
(D) Ag_2SO_3 , $K_{\text{sp}} = 1.5 \times 10^{-14}$
35. What is the pH of a 0.10 M solution of NH_4F ? The K_{a} of NH_4^+ is 5.6×10^{-10} and the K_{a} of HF is 6.8×10^{-4} .
- (A) 2.08 (B) 5.12 (C) 6.21 (D) 8.08

36. 1.000 g of which salt, when titrated with 0.500 M HCl , gives the titration curve shown?



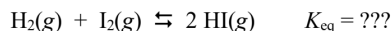
- (A) Potassium cyanide, KCN ($M = 65.12$, $\text{p}K_{\text{a}}$ of $\text{HCN} = 9.2$)
(B) Sodium nitrite, NaNO_2 ($M = 69.00$, $\text{p}K_{\text{a}}$ of $\text{HNO}_2 = 3.4$)
(C) Potassium phenoxide, $\text{K}(\text{C}_6\text{H}_5\text{O})$ ($M = 116.20$, $\text{p}K_{\text{a}}$ of $\text{C}_6\text{H}_5\text{OH} = 9.9$)
(D) Sodium pivalate, $\text{Na}(\text{C}_4\text{H}_9\text{COO})$ ($M = 124.11$, $\text{p}K_{\text{a}}$ of $\text{C}_4\text{H}_9\text{COOH} = 4.9$)
37. What is the correct equilibrium expression for the given reaction?
- $$\text{Cu}(\text{OH})_2(s) + 4 \text{NH}_3(aq) \rightleftharpoons \text{Cu}(\text{NH}_3)_4^{2+}(aq) + 2 \text{OH}^-(aq)$$
- (A) $K_{\text{eq}} = \frac{[\text{Cu}(\text{NH}_3)_4^{2+}][\text{OH}^-]}{[\text{NH}_3]}$
(B) $K_{\text{eq}} = \frac{[\text{Cu}(\text{NH}_3)_4^{2+}][2 \text{OH}^-]}{[4 \text{NH}_3]}$
(C) $K_{\text{eq}} = \frac{[\text{Cu}(\text{NH}_3)_4^{2+}][\text{OH}^-]^2}{[\text{NH}_3]^4}$
(D) $K_{\text{eq}} = \frac{[\text{Cu}(\text{NH}_3)_4^{2+}][2 \text{OH}^-]^2}{[4 \text{NH}_3]^4}$
38. 0.100 mol of HF ($K_{\text{a}} = 6.6 \times 10^{-4}$) is added to water to make 1.00 L of solution. Which statement is correct at equilibrium?
- (A) $[\text{H}_3\text{O}^+] = [\text{HF}]$ (B) $[\text{H}_3\text{O}^+] = 0.100 \text{ M}$
(C) $[\text{HF}] > [\text{F}^-]$ (D) $[\text{H}_3\text{O}^+] > [\text{HF}]$
39. What is the solubility of silver oxalate, $\text{Ag}_2\text{C}_2\text{O}_4$ ($K_{\text{sp}} = 3.5 \times 10^{-11}$), in a 0.050 M sodium oxalate solution?
- (A) $1.4 \times 10^{-8} \text{ M}$ (B) $5.9 \times 10^{-6} \text{ M}$
(C) $1.3 \times 10^{-5} \text{ M}$ (D) $2.1 \times 10^{-4} \text{ M}$

Equilibrium

40. A pure sample of a monoprotic acid is dissolved in water. The sample is titrated with sodium hydroxide solution. At the point where 20.0 mL of the NaOH solution has been added, the pH is 4.15. The phenolphthalein endpoint of the titration is observed when 50.0 mL of NaOH have been added. What is the pK_a of the acid?

(A) 4.15 (B) 4.33 (C) 4.55 (D) 5.19

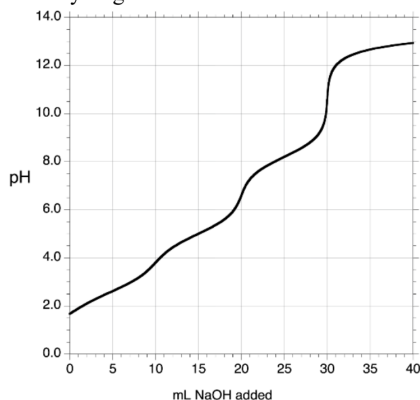
41. Consider the reaction:



Into a 1.00 L vessel, 1.00 mol $H_2(g)$ and 1.00 mol $I_2(g)$ are placed at a high temperature. When the reaction mixture stops changing, it is found that 79.0% of the $H_2(g)$ has reacted. What is the equilibrium constant for this reaction at this temperature?

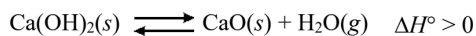
(A) 14.2 (B) 17.9 (C) 35.8 (D) 56.6

42. A triprotic acid is titrated with sodium hydroxide to give the titration curve below. What are the pK_a values of the three acidic hydrogens?



(A) 2.6, 5.0, 8.2 (B) 3.8, 6.6, 10.6
(C) 3.8, 8.2, 12.7 (D) 5.0, 8.2, 12.7

43. The endothermic reaction shown below is at equilibrium in a sealed flask, with significant amounts of both $Ca(OH)_2(s)$ and $CaO(s)$ present. Which action will increase the amount of $Ca(OH)_2(s)$ at equilibrium?

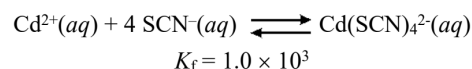


(A) Crushing the $Ca(OH)_2(s)$ into smaller pieces
(B) Decreasing the temperature of the flask
(C) Adding more $CaO(s)$ to the flask
(D) Adding $N_2(g)$ to the flask

44. A solution of ammonia, NH_3 , has $pH = 11.50$. What is the ammonia concentration? (The pK_a of NH_4^+ is 9.24.)

(A) $1.7 \times 10^{-5} M$ (B) $3.2 \times 10^{-3} M$
(C) $5.5 \times 10^{-3} M$ (D) 0.58 M

45. At equilibrium, a solution contains 0.50 mol each of $Cd^{2+}(aq)$, $SCN^-(aq)$, and $Cd(SCN)_4^{2-}(aq)$. What is the solution's volume?

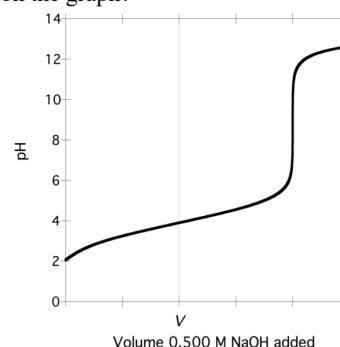


(A) 2.0 L (B) 2.2 L (C) 2.8 L (D) 8.0 L

46. Barium fluoride has $K_{sp} = 1.8 \times 10^{-7}$. What is the maximum fluoride ion concentration possible in a solution with $[Ba^{2+}] = 5.0 \times 10^{-4} M$?

(A) $3.6 \times 10^{-4} M$ (B) $3.6 \times 10^{-3} M$
(C) $9.5 \times 10^{-3} M$ (D) $1.9 \times 10^{-2} M$

47. Trimesic acid is a triprotic acid with $pK_1 = 3.1$, $pK_2 = 3.9$, and $pK_3 = 4.7$. 100.0 mL of 0.100 M trimesic acid is titrated with 0.500 M NaOH(aq) to give the graph below. What volume of NaOH(aq) has been added at the point marked V on the graph?



(A) 5 mL (B) 10 mL (C) 20 mL (D) 30 mL

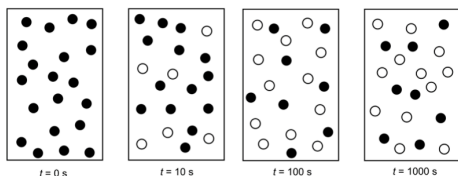
48. A solution containing $Co(H_2O)_6^{2+}(aq)$, $CoCl_4^{2-}(aq)$, and $Cl^-(aq)$ at equilibrium at room temperature is initially pink. When heated, the solution turns blue. Then, when $Ag^+(aq)$ is added, the solution turns back to pink. Which statements are correct?

I. $Co(H_2O)_6^{2+}(aq)$ is pink.
II. Formation of $CoCl_4^{2-}(aq)$ from $Co(H_2O)_6^{2+}(aq)$ and $Cl^-(aq)$ is exothermic.

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

Equilibrium

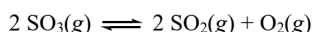
49. The diagrams below represent the progress of a reaction of $A \rightarrow B$, with molecules of A represented by solid circles and molecules of B represented by open circles. Which statement best describes the reaction at $t = 1000$ s?



- (A) The reaction is at equilibrium because the concentrations of A and B are no longer changing.
 (B) The reaction is not at equilibrium because not all of the A has been consumed.
 (C) The reaction is not at equilibrium because the concentrations of A and B are not equal.
 (D) The reaction is not at equilibrium because the positions of the molecules are still changing.
50. What is the pH of a 0.10 M solution of NaCN? The K_a of HCN is 4.9×10^{-10} .
 (A) 2.85 (B) 5.15 (C) 8.85 (D) 11.15
51. For which of the following reactions would the yield of products at equilibrium NOT increase at a higher pressure?
 (A) $N_2(g) + O_2(g) \rightleftharpoons 2 NO(g)$
 (B) $Ti(s) + 2 Cl_2(g) \rightleftharpoons TiCl_4(g)$
 (C) $2 C_2H_4(g) + 2 H_2O(g) \rightleftharpoons 2 C_2H_6(g) + O_2(g)$
 (D) $4 HCl(g) + O_2(g) \rightleftharpoons 2 H_2O(l) + 2 Cl_2(g)$
52. Chromium(III) hydroxide has $K_{sp} = 1.6 \times 10^{-30}$. What is the molar solubility of $Cr(OH)_3$ in a solution whose pH is maintained at 6.00?
 (A) $1.6 \times 10^{-12} \text{ mol L}^{-1}$ (B) $1.6 \times 10^{-8} \text{ mol L}^{-1}$
 (C) $3.6 \times 10^{-8} \text{ mol L}^{-1}$ (D) $1.6 \times 10^{-6} \text{ mol L}^{-1}$
53. At 400 K, this reaction has $K_p = 8.2 \times 10^{-4}$.

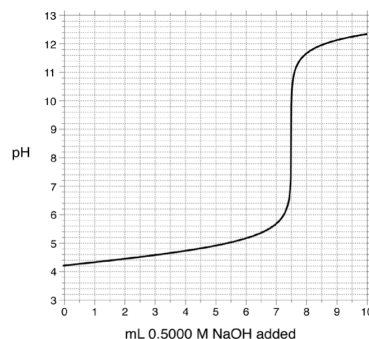


What is K_p at 400 K for the following reaction?



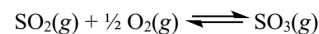
- (A) 6.7×10^{-7} (B) 8.2×10^{-4}
 (C) 1.6×10^{-3} (D) 2.9×10^{-2}

54. An unknown monoprotic carboxylic acid RCOOH is mixed in a 1:1 mole ratio with its sodium salt, Na(RCOO). A 1.000 g sample of this 1:1 mixture is dissolved in 50 mL water and titrated with 0.5000 M NaOH solution. The pH is measured as a function of added titrant to give the following titration curve:



Which statements are correct?

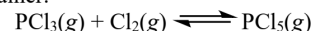
- I. The molar mass of RCOOH is within 5% of 265 g mol^{-1} .
 II. The pK_a of RCOOH is within 0.1 unit of 4.2.
- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II
55. What mass of silver chloride ($M = 143.4$) will dissolve in 1.00 L of water? The K_{sp} of AgCl is 1.8×10^{-10} .
 (A) 1.4 mg (B) 1.9 mg (C) 2.9 mg (D) 3.8 mg
56. What is the pH of a 0.20 M solution of sodium benzoate, Na(C₆H₅COO)? The K_a of benzoic acid, C₆H₅COOH, is 6.5×10^{-5} .
 (A) 5.26 (B) 8.74 (C) 9.09 (D) 11.56
57. Sulfur trioxide is formed from the reaction of sulfur dioxide and oxygen:



At 1000 K, an equilibrium mixture has partial pressures of 0.562 bar SO_2 , 0.101 bar O_2 , and 0.332 bar SO_3 . What is the equilibrium constant K_p for the reaction at this temperature?

- (A) 1.86 (B) 3.46 (C) 5.85 (D) 16.8

58. The following endothermic reaction is at equilibrium in a sealed container.



Which of the following changes would result in an increase in the number of moles of $PCl_5(g)$ present at equilibrium?

- I. Increasing the temperature II. Increasing the volume
 (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

Equilibrium

59. A 0.12 M solution of a monoprotic acid is 2.3% ionized. What is the K_a of this acid?

(A) 2.8×10^{-3} (B) 3.3×10^{-4}
(C) 6.5×10^{-5} (D) 7.6×10^{-6}

60. Calcium fluoride, CaF_2 , has a molar solubility of $2.1 \times 10^{-4} \text{ mol L}^{-1}$ at $\text{pH} = 7.00$. By what factor does its molar solubility increase in a solution with $\text{pH} = 3.00$? The $\text{p}K_a$ of HF is 3.17.

(A) 1.48 (B) 1.83 (C) 2.48 (D) 4.96

61. Which solution has the greatest percent ionization?

(A) 0.010 M formic acid ($K_a = 1.8 \times 10^{-4}$)
(B) 0.10 M formic acid ($K_a = 1.8 \times 10^{-4}$)
(C) 0.010 M acetic acid ($K_a = 1.8 \times 10^{-5}$)
(D) 0.10 M acetic acid ($K_a = 1.8 \times 10^{-5}$)

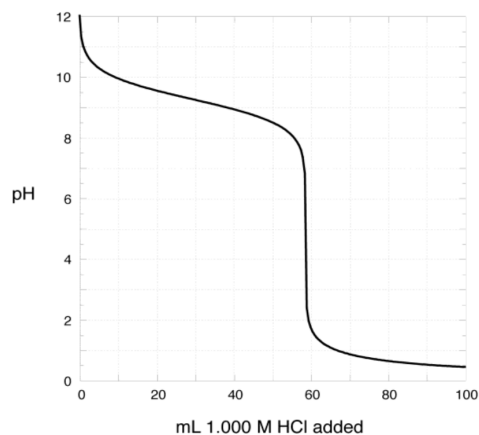
62. Calcium oxalate, CaC_2O_4 ($M = 128.1$), dissolves to the extent of 0.67 mg L^{-1} . What is its K_{sp} ?

(A) 6.7×10^{-4} (B) 4.5×10^{-7}
(C) 2.7×10^{-11} (D) 5.7×10^{-16}

63. What is the concentration of a solution of K_2CO_3 that has $\text{pH} = 11.90$? (For H_2CO_3 , $K_{a1} = 4.2 \times 10^{-7}$, $K_{a2} = 4.8 \times 10^{-11}$.)

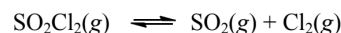
(A) $3.0 \times 10^{-1} \text{ M}$ (B) $2.6 \times 10^{-2} \text{ M}$
(C) $7.9 \times 10^{-3} \text{ M}$ (D) $1.3 \times 10^{-12} \text{ M}$

64. 1.000 g of a weak base is titrated with 1.000 M aqueous HCl to give the data shown. What is the identity of the base?



- (A) Ammonia, NH_3 ($\text{p}K_a$ of $\text{NH}_4^+ = 9.3$)
(B) Aniline, $\text{C}_6\text{H}_5\text{NH}_2$ ($\text{p}K_a$ of $\text{C}_6\text{H}_5\text{NH}_3^+ = 4.6$)
(C) Hydroxylamine, NH_2OH ($\text{p}K_a$ of $\text{NH}_3\text{OH}^+ = 6.0$)
(D) Trimethylamine, $(\text{CH}_3)_3\text{N}$ ($\text{p}K_a$ of $(\text{CH}_3)_3\text{NH}^+ = 9.8$)

65. Sulfuryl chloride is in equilibrium with sulfur dioxide and chlorine gas:



A system with a volume of 1.00 L is in equilibrium at a certain temperature with $p(\text{SO}_2\text{Cl}_2) = 1.00 \text{ bar}$ and $p(\text{SO}_2) = p(\text{Cl}_2) = 0.10 \text{ bar}$. By how much will the number of moles of SO_2Cl_2 at equilibrium change if the volume is reduced to 0.50 L?

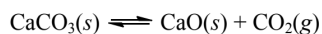
- (A) Increase 1-10% (B) Increase 11-50%
(C) Decrease 1-10% (D) Decrease 11-50%

66. Which of the following will increase $[\text{Cl}^-]$ in a saturated solution of AgCl in contact with excess solid silver chloride?

- I. Addition of $\text{AgCl}(s)$ II. Addition of $\text{NH}_3(aq)$
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

Equilibrium

67. A 60.0 g sample of CaCO_3 is heated to 950 K in a 1.00 L evacuated container, where it reacts according to the following equation:



After equilibrium is attained, the pressure of $\text{CO}_2(g)$ is 30.0 mm Hg. When the experiment is repeated using 120.0 g CaCO_3 , what is the equilibrium pressure P ?

- (A) 15.0 mm Hg $\leq P < 30.0$ mm Hg
(B) $P = 30.0$ mm Hg
(C) 30.0 mm Hg $< P < 60.0$ mm Hg
(D) $P = 60.0$ mm Hg
68. Nitrous acid, HNO_2 , has $K_a = 4.5 \times 10^{-4}$. What is the best description of the species present in a 0.1 M solution of nitrous acid?
- (A) $\text{HNO}_2(aq)$ is the predominant species; much smaller amounts of $\text{H}^+(aq)$ and $\text{NO}_2^-(aq)$ exist.
(B) $\text{H}^+(aq)$ and $\text{NO}_2^-(aq)$ are the predominant species; much smaller amounts of $\text{HNO}_2(aq)$ exist.
(C) Only $\text{H}^+(aq)$ and $\text{NO}_2^-(aq)$ are present in measurable amounts.
(D) $\text{HNO}_2(aq)$, $\text{H}^+(aq)$, and $\text{NO}_2^-(aq)$ are all present in comparable amounts.
69. Silver sulfate, Ag_2SO_4 ($M = 311.87$), has $K_{sp} = 6.0 \times 10^{-5}$. What mass of Ag_2SO_4 will dissolve per liter of water?
- (A) 0.019 g (B) 1.7 g
(C) 2.4 g (D) 7.7 g
70. $\text{IBr}(g)$ is in equilibrium with $\text{I}_2(g)$ and $\text{Br}_2(g)$ at 150 °C:
- $$2 \text{IBr}(g) \rightleftharpoons \text{I}_2(g) + \text{Br}_2(g) \quad K = 8.50 \times 10^{-3}$$
- Initially, a closed vessel at 150 °C has a partial pressure of IBr of 0.350 atm and partial pressures of I_2 and Br_2 each of 0.750 atm. What is the partial pressure of IBr once the system reaches equilibrium?
- (A) 1.01 atm (B) 1.09 atm
(C) 1.56 atm (D) 1.82 atm
71. 20 mL of an approximately 10% aqueous solution of ethylamine, $\text{CH}_3\text{CH}_2\text{NH}_2$, is titrated with 0.3000 M aqueous HCl . Which indicator would be most suitable for this titration? The $\text{p}K_a$ of $\text{CH}_3\text{CH}_2\text{NH}_3^+$ is 10.75.
- (A) Thymol blue, color change from pH = 1.2 to 2.8
(B) Bromocresol green, color change from pH = 4.0 to 5.6
(C) Phenolphthalein, color change from pH = 8.0 to 10.0
(D) Alizarin yellow R, color change from pH = 10.0 to 12.0

72. The tetramminecopper(II) ion, $\text{Cu}(\text{NH}_3)_4^{2+}$, has a formation constant $K_f = 1.1 \times 10^{13}$. What is the minimum concentration of free ammonia in solution required to ensure that at least 99.9% of the dissolved copper(II) ion is found in the form of its ammonia complex?

- (A) 9×10^{-14} M (B) 9×10^{-11} M
(C) 8×10^{-4} M (D) 3×10^{-3} M

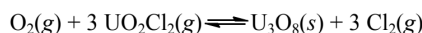
73. When 0.10 M solutions of ammonium acetate, barium acetate, and sodium acetate are ranked from least basic to most basic, what is the correct ordering?

- (A) $\text{NH}_4\text{C}_2\text{H}_3\text{O}_2 < \text{NaC}_2\text{H}_3\text{O}_2 < \text{Ba}(\text{C}_2\text{H}_3\text{O}_2)_2$
(B) $\text{Ba}(\text{C}_2\text{H}_3\text{O}_2)_2 < \text{NH}_4\text{C}_2\text{H}_3\text{O}_2 < \text{NaC}_2\text{H}_3\text{O}_2$
(C) $\text{NaC}_2\text{H}_3\text{O}_2 < \text{Ba}(\text{C}_2\text{H}_3\text{O}_2)_2 < \text{NH}_4\text{C}_2\text{H}_3\text{O}_2$
(D) $\text{NaC}_2\text{H}_3\text{O}_2 < \text{NH}_4\text{C}_2\text{H}_3\text{O}_2 < \text{Ba}(\text{C}_2\text{H}_3\text{O}_2)_2$

74. What is the solubility of MgF_2 ($K_{sp} = 6.8 \times 10^{-9}$) in pure water?

- (A) $6.8 \times 10^{-9} \text{ mol} \cdot \text{L}^{-1}$ (B) $5.8 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1}$
(C) $8.2 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1}$ (D) $1.2 \times 10^{-3} \text{ mol} \cdot \text{L}^{-1}$

75. What is the ratio K_c/K_p for the following reaction at 723 °C?

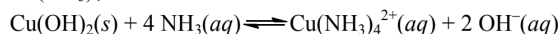


- (A) 0.0122 (B) 1.00 (C) 59.4 (D) 81.8

76. What is $[\text{H}_3\text{O}^+]$ in a solution formed by dissolving 1.00 g NH_4Cl ($M = 53.5$) in 30.0 mL of 3.00 M NH_3 ($K_b = 1.8 \times 10^{-5}$)?

- (A) 2.7×10^{-9} M (B) 5.5×10^{-10} M
(C) 1.2×10^{-10} M (D) 1.4×10^{-12} M

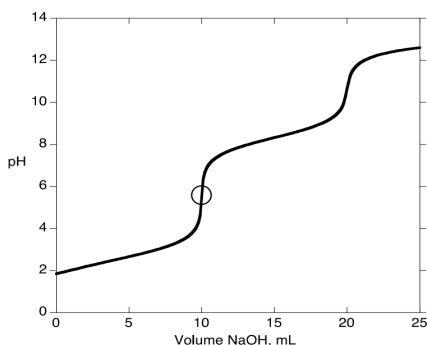
77. Copper(II) hydroxide, $\text{Cu}(\text{OH})_2$, has $K_{sp} = 2.2 \times 10^{-20}$. For the reaction below, $K_{eq} = 4.0 \times 10^{-7}$. What is K_f for $\text{Cu}(\text{NH}_3)_4^{2+}$?



- (A) 8.8×10^{-27} (B) 5.5×10^{-14}
(C) 1.8×10^{13} (D) 1.1×10^{26}

Equilibrium

78. A 0.100 M aqueous solution of H_2SeO_3 is titrated with 1.000 M NaOH solution. At the point marked with a circle on the titration curve, which species represent at least 10% of the total selenium in solution?



- (A) H_2SeO_3 only (B) Both H_2SeO_3 and HSeO_3^-
 (C) HSeO_3^- only (D) Both HSeO_3^- and SeO_3^{2-}

Questions 79 and 80 should be answered with reference to the equation



79. What is the equilibrium expression for the reaction represented by the equation above?

- (A) $K = \frac{2[\text{CO}]}{2[\text{C}] + [\text{O}_2]}$ (B) $K = \frac{2[\text{CO}]}{[\text{O}_2]}$
 (C) $K = \frac{[\text{CO}]^2}{[\text{C}]^2[\text{O}_2]}$ (D) $K = \frac{[\text{CO}]^2}{[\text{O}_2]}$

80. If the reaction is at equilibrium with excess C(s) remaining, what change will increase the quantity of CO(g) for the reaction at equilibrium?

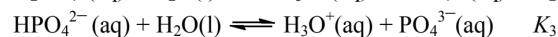
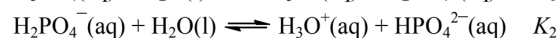
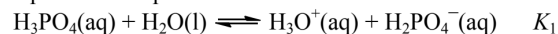
I Adding C(s)
 II Increasing the temperature
 III Increasing the pressure

- (A) I only (B) III only
 (C) I, II, and III (D) None of these

81. An endothermic reaction has a positive value for ΔS . Which of the following is true about the equilibrium constant for this reaction?

- (A) It may be greater than 1 only at low temperatures.
 (B) It may be greater than 1 only at high temperatures.
 (C) It is greater than 1 at all temperatures.
 (D) It is less than 1 at all temperatures.

82. Phosphoric acid, H_3PO_4 , establishes the following equilibria in aqueous solution.



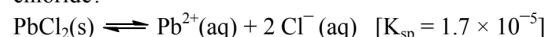
What is the relationship among the K values?

- (A) $K_1 = K_2 = K_3$ (B) $K_1 > K_2 = K_3$
 (C) $K_1 > K_2 > K_3$ (D) $K_1 < K_2 < K_3$

83. Which pair of solutes could be used to prepare an aqueous buffer solution with a pH < 7?

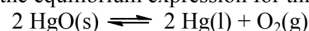
- (A) $\text{HCl} - \text{NH}_4\text{Cl}$ (B) $\text{HF} - \text{NaF}$
 (C) $\text{NH}_3 - \text{NH}_4\text{Cl}$ (D) $\text{NaOH} - \text{NaCl}$

84. What will be the result when 15.0 mL of 0.040 M lead(II) nitrate is mixed with 15.0 mL of 0.040 M sodium chloride?



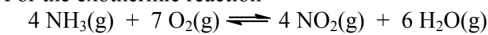
- (A) A clear solution with no precipitate will result.
 (B) Solid PbCl_2 will precipitate and excess Pb^{2+} ions will remain in solution.
 (C) Solid PbCl_2 will precipitate and excess Cl^- ions will remain in solution.
 (D) Solid PbCl_2 will precipitate and no excess ions will remain in solution.

85. What is the equilibrium expression for this reaction?



- (A) $K = [\text{Hg}][\text{O}_2]/[\text{HgO}]$
 (B) $K = [\text{Hg}]^2[\text{O}_2]/[\text{HgO}]^2$
 (C) $K = [\text{Hg}]^2[\text{O}_2]$
 (D) $K = [\text{O}_2]$

86. For the exothermic reaction



which change will increase the quantity of NO_2 in the mixture?

- (A) increasing temperature
 (B) decreasing container volume
 (C) adding Ne(g)
 (D) adding $\text{H}_2\text{O(g)}$

87. Weak acids include which of the following?

I. HF(aq) II. HI(aq) III. $\text{HNO}_2\text{(aq)}$

- (A) I. only (B) II. only
 (C) III. only (D) I. and III. only

Equilibrium

88. $\text{H}_2\text{CO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{HCO}_3^-(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
 $\text{HCO}_3^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{CO}_3^{2-}(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
 According to the equations above, what is the conjugate base of HCO_3^- ?
- (A) $\text{H}_2\text{CO}_3(\text{aq})$ (B) $\text{H}_2\text{O}(\text{l})$
 (C) $\text{H}_3\text{O}^+(\text{aq})$ (D) $\text{CO}_3^{2-}(\text{aq})$
89. A saturated solution of which salt will have the highest $[\text{Ag}^+]$?
- (A) AgCl ($K_{\text{sp}} = 1.8 \times 10^{-10}$)
 (B) Ag_2CrO_4 ($K_{\text{sp}} = 1.1 \times 10^{-12}$)
 (C) Ag_3PO_4 ($K_{\text{sp}} = 1.8 \times 10^{-18}$)
 (D) Ag_2S ($K_{\text{sp}} = 6.0 \times 10^{-51}$)
90. A saturated solution of manganese(II) carbonate ($M = 114.95$) contains 5.44×10^{-5} g of MnCO_3 per 100 mL at 25°C . What is its K_{sp} at this temperature?
- (A) 4.7×10^{-6} (B) 3.0×10^{-9}
 (C) 2.2×10^{-11} (D) 2.2×10^{-13}
91. Which statement(s) characterize(s) a chemical system at equilibrium?
- I The rate of the forward reaction is equal to the rate of the reverse reaction.
 II The concentrations of the reactants and products are equal.
- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II
92. What is the solubility in pure water of $\text{Ba}(\text{IO}_3)_2$ in moles per liter at 25°C ? [$K_{\text{sp}}(25^\circ\text{C}) = 6.0 \times 10^{-10}$]
- (A) 1.2×10^{-5} (B) 1.7×10^{-5}
 (C) 5.3×10^{-4} (D) 8.4×10^{-4}
93. $\text{C}(\text{s}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + \text{H}_2(\text{g}) \quad \Delta H > 0$
 For the system above at equilibrium, which changes will increase the amount of $\text{H}_2(\text{g})$?
- I Adding $\text{C}(\text{s})$
 II Increasing the volume of the container
 III Increasing the temperature
- (A) I only (B) III only
 (C) II and III only (D) I, II and III
94. Determine the volume of 0.125 M NaOH required to titrate to the equivalence point 25.0 mL of a 0.175 M solution of a monoprotic weak acid that is 20% ionized.
- (A) 7.00 mL (B) 17.9 mL
 (C) 28.0 mL (D) 35.0 mL
95. A 75 mL solution that is 0.10 M in $\text{HC}_2\text{H}_3\text{O}_2$ and 0.10 M in $\text{NaC}_2\text{H}_3\text{O}_2$ has a pH of 4.74. Which of the following actions will change the pH of this solution?
- I Adding 15 mL of 0.10 M HCl
 II Adding 0.010 mol of $\text{NaC}_2\text{H}_3\text{O}_2$
 III Diluting the solution from 75 mL to 125 mL
- (A) I only (B) II only
 (C) I and II only (D) I, II and III
96. Which acid-base indicator will give the best results for the titration of an aqueous ammonia solution with 0.10 M HCl?
- | Indicator | Color change | pH range |
|-----------------|------------------------------|--------------------|
| Methyl violet | red $\rightarrow$ blue | 0 $\rightarrow$ 2 |
| Methyl red | red $\rightarrow$ yellow | 4 $\rightarrow$ 6 |
| Cresol red | yellow $\rightarrow$ purple | 7 $\rightarrow$ 9 |
| Phenolphthalein | colorless $\rightarrow$ pink | 8 $\rightarrow$ 10 |
- (A) Methyl violet (B) Methyl red
 (C) Cresol red (D) Phenolphthalein
97. For the equilibrium system:
 $\text{CO}(\text{g}) + 2 \text{H}_2(\text{g}) \rightleftharpoons \text{CH}_3\text{OH}(\text{l})$
 what is K_{c} ?
- (A) $K_{\text{c}} = \frac{[\text{CH}_3\text{OH}]}{2[\text{CO}][\text{H}_2]}$ (B) $K_{\text{c}} = \frac{[\text{CH}_3\text{OH}]}{[\text{CO}][\text{H}_2]^2}$
 (C) $K_{\text{c}} = \frac{1}{2[\text{CO}][\text{H}_2]}$ (D) $K_{\text{c}} = \frac{1}{[\text{CO}][\text{H}_2]^2}$
98. Consider the system at equilibrium:
 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$
 for which $\Delta H < 0$. Which change(s) will increase the yield of $\text{SO}_3(\text{g})$?
- I Increasing the temperature
 II Increasing the volume of the container
- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II
99. When 0.10 M solutions of HF, HCl, KF, and KCl are arranged in order of increasing pH which order is correct?
- (A) HF, HCl, KF, KCl (B) HCl, HF, KF, KCl
 (C) HCl, HF, KCl, KF (D) HF, HCl, KCl, KF
100. Which one of the following mixtures creates a buffer solution?
- (A) 10.0 mL 0.10 M NaOH, 10.0 mL 0.10 M HF
 (B) 20.0 mL 0.10 M NaOH, 15.0 mL 0.10 M HF
 (C) 15.0 mL 0.10 M NaOH, 20.0 mL 0.10 M HF
 (D) 10.0 mL 0.10 M NaOH, 5.0 mL 0.20 M HF

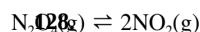
Equilibrium

- 101.** The solubility of AgBrO_3 in aqueous solution depends on the presence of other substances in solution. Relative to its solubility in H_2O the solubility of AgBrO_3 is higher in 0.10 M _____ and lower in 0.10 M _____.
- (A) NH_3 , KBrO_3 (B) KBrO_3 , NH_3
 (C) HNO_3 , NH_3 (D) NH_3 , HNO_3
- 102.** A 0.015 M solution of a weak acid has a pH of 3.52. What is the value of the K_a for this acid?
- (A) 2.0×10^{-2} (B) 6.2×10^{-6}
 (C) 9.1×10^{-8} (D) 1.4×10^{-9}
- 103.** What is the equilibrium expression, K_c , for the reaction:
 $2\text{S}(\text{s}) + 3\text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$
- (A) $K_c = 2[\text{SO}_3]/(2[\text{S}] + 3[\text{O}_2])$ (B) $K_c = 2[\text{SO}_3]/3[\text{O}_2]$
 (C) $K_c = [\text{SO}_3]^2/[\text{S}]^2[\text{O}_2]^3$ (D) $K_c = [\text{SO}_3]^2/[\text{O}_2]^3$
- 104.** In which reaction at equilibrium will the amount of reactants present increase with an increase in the container volume?
- (A) $\text{C}(\text{s}) + \text{CO}_2(\text{g}) \rightleftharpoons 2\text{CO}(\text{g})$
 (B) $\text{H}_2(\text{g}) + \text{F}_2(\text{g}) \rightleftharpoons 2\text{HF}(\text{g})$
 (C) $\text{CO}(\text{g}) + \text{NO}_2(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{NO}(\text{g})$
 (D) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
- 105.** What is the pH of a solution made by mixing 200. mL of 0.0657 M NaOH , 140. mL of 0.107 M HCl , and 160. mL of H_2O ?
- (A) 3.04 (B) 2.74 (C) 2.43 (D) 2.27
- 106.** What is the percentage ionization of HCOOH molecules in a 0.10 M solution? [$K_a = 1.8 \times 10^{-4}$]
- (A) 4.2% (B) 2.7% (C) 1.8% (D) 1.3%
- 107.** What is the pH of a solution that is 0.20 M in HF and 0.40 M in NaF ? [$K_a = 7.2 \times 10^{-4}$]
- (A) 1.92 (B) 2.84 (C) 3.14 (D) 3.44
- 108.** A 500. mL saturated solution of MgCO_3 ($M = 84$) is reduced to 120. mL by evaporation. What mass of solid MgCO_3 is formed? [$K_{sp} = 4.0 \times 10^{-5}$]
- (A) 0.0013 g (B) 0.064 g
 (C) 0.20 g (D) 0.27 g
- 109.** What is the K_{eq} expression for the reaction,
 $\text{C}(\text{s}) + \text{CO}_2(\text{g}) \rightleftharpoons 2\text{CO}(\text{g})$?
- (A) $K_{eq} = \frac{2[\text{CO}]}{[\text{CO}_2]}$ (B) $K_{eq} = \frac{2[\text{C}][\text{CO}]}{[\text{CO}_2]}$
 (C) $K_{eq} = \frac{[\text{CO}]^2}{[\text{CO}_2]}$ (D) $K_{eq} = \frac{[\text{C}][\text{CO}]^2}{[\text{CO}_2]}$
- 110.** The equilibrium system $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ has $K_p = 11$ and $\Delta H^\circ = 57 \text{ kJ}\cdot\text{mol}^{-1}$ at 25°C . Which action will **not** cause a change in the position of the equilibrium?
- (A) increasing the temperature
 (B) adding $\text{NO}_2(\text{g})$
 (C) adding xenon gas to increase the pressure
 (D) increasing the container volume
- 111.** Which is not a conjugate acid/base pair?
- (A) H_2CO_3 & CO_3^{2-} (B) HSO_4^- & SO_4^{2-}
 (C) H_2PO_4^- & HPO_4^{2-} (D) H_3O^+ & H_2O
- 112.** What is the $[\text{OH}^-]$ in an aqueous solution which has a pH = 11.70?
- (A) $7.1 \times 10^{-2} \text{ M}$ (B) $5.0 \times 10^{-3} \text{ M}$
 (C) $1.4 \times 10^{-6} \text{ M}$ (D) $2.0 \times 10^{-12} \text{ M}$
- 113.** Equal volumes of 0.25 M HNO_2 and 0.25 M HNO_3 are titrated separately with 0.25 M KOH . Which would be the same for both titrations?
- (A) initial pH
 (B) pH halfway to the equivalence point
 (C) pH at the equivalence point
 (D) pH when 5 mL excess KOH has been added
- 114.** For which salt is the molar solubility, s , equal to $4 \times 10^{-6} \text{ M}$?
- (A) $\text{AgC}_2\text{H}_3\text{O}_2$ $K_{sp} = 2 \times 10^{-3}$
 (B) TlBr $K_{sp} = 4 \times 10^{-6}$
 (C) MnCO_3 $K_{sp} = 2 \times 10^{-11}$
 (D) $\text{Zn}(\text{OH})_2$ $K_{sp} = 3 \times 10^{-17}$

Equilibrium

115. Which statement is correct about a system at equilibrium?
- (A) The forward and reverse reactions occur at identical rates.
 (B) The concentrations of reactants must equal the concentrations of the products.
 (C) The concentrations of reactants and products can be changed by adding a catalyst.
 (D) The concentrations of reactants and products are not affected by a change in temperature.
116. For which reaction at equilibrium will an decrease in volume at constant temperature cause a decrease in the amount of product?
- (A) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
 (B) $\text{HCl}(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{Cl}^-(\text{aq})$
 (C) $\text{Fe}_3\text{O}_4(\text{s}) + 4\text{H}_2(\text{g}) \rightleftharpoons 3\text{Fe}(\text{s}) + 4\text{H}_2\text{O}(\text{g})$
 (D) $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
117. What is the conjugate base of HSO_4^- ?
- (A) H^+ (B) H_2SO_4 (C) OH^- (D) SO_4^{2-}
118. The K_a of hydrocyanic acid, HCN, is 5.0×10^{-10} . What is the pH of 0.050 M HCN(aq)?
- (A) between 3.5 and 4.5 (B) between 5.0 and 5.5
 (C) between 9.0 and 9.5 (D) between 10.5 and 11.0
119. Which pair of solutions forms a buffer solution when equal volumes of each are mixed?
- (A) 0.20 M HCl and 0.20 M NaOH
 (B) 0.40 M $\text{HC}_2\text{H}_3\text{O}_2$ and 0.20 M NaOH
 (C) 0.20 M HCl and 0.20 M NH_3
 (D) 0.40 M HCl and 0.20 M NH_3
120. Which statement is correct about the initial precipitate that forms when a 0.10 M NaF solution is added slowly to a solution that is 0.10 M with respect to both $\text{Ca}(\text{NO}_3)_2$ and $\text{Mg}(\text{NO}_3)_2$?
- | | K_{sp} |
|----------------|-----------------------|
| CaF_2 | 4.0×10^{-11} |
| MgF_2 | 6.4×10^{-9} |
- (A) The initial precipitate will contain CaF_2 only.
 (B) The initial precipitate will contain MgF_2 only.
 (C) The initial precipitate will contain both CaF_2 and MgF_2 with more CaF_2 .
 (D) The initial precipitate will contain both CaF_2 and MgF_2 with more MgF_2 .
121. For which reaction at equilibrium does a decrease in volume of the container cause a decrease in product(s) at constant temperature?
- (A) $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
 (B) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$
 (C) $\text{HCl}(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{Cl}^-(\text{aq})$
 (D) $\text{SO}_2(\text{g}) + \text{NO}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g}) + \text{NO}(\text{g})$
122. Which is the weakest acid?
- (A) ascorbic acid ($K_a = 8.0 \times 10^{-5}$)
 (B) boric acid ($K_a = 5.8 \times 10^{-10}$)
 (C) butyric acid ($K_a = 1.5 \times 10^{-5}$)
 (D) hydrocyanic acid ($K_a = 4.9 \times 10^{-10}$)
123. At 20.0 °C water has $K_w = 6.807 \times 10^{-15}$. What is the pH of pure water at this temperature?
- (A) 6.667 (B) 6.920 (C) 7.000 (D) 7.084
124. Which solution has the highest pH?
- | | K_a |
|--------------------------|-----------------------|
| CH_3COOH | 1.8×10^{-5} |
| HCN | 5.8×10^{-10} |
- (A) 0.10 M CH_3COOH (B) 0.10 M HCN
 (C) 0.10 M CH_3COOK (D) 0.10 M NaBr
125. What happens to the pH of a buffer solution when it is diluted by a factor of 10?
- (A) The buffer pH decreases by 1 unit.
 (B) The buffer pH increases by 1 unit.
 (C) The change in pH depends on the buffer used.
 (D) The pH does not change appreciably.
126. The solubility of PbI_2 is $1.3 \times 10^{-3} \text{ mol} \cdot \text{L}^{-1}$. What is the K_{sp} for PbI_2 ?
- (A) 2.2×10^{-9} (B) 8.8×10^{-9}
 (C) 1.7×10^{-6} (D) 3.4×10^{-6}
127. $2\text{NO}_2(\text{g}) + 7\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}) + 4\text{H}_2\text{O}(\text{l})$
 What is the correct equilibrium expression for this reaction?
- (A) $K_c = \frac{[\text{NH}_3]^2}{[\text{NO}_2]^2[\text{H}_2]^7}$ (B) $K_c = \frac{[\text{NO}_2]^2[\text{H}_2]^7}{[\text{NH}_3]^2}$
 (C) $K_c = \frac{[\text{NH}_3]^2[\text{H}_2\text{O}]^4}{[\text{NO}_2]^2[\text{H}_2]^7}$ (D) $K_c = \frac{[\text{NH}_3]^2[\text{H}_2\text{O}]^4}{[\text{NO}_2]^2}$

Equilibrium



The equilibrium reaction shown is endothermic as written. Which change will increase the amount of NO_2 at equilibrium?

- (A) adding a catalyst
(B) decreasing the temperature
(C) increasing the volume of the container
(D) adding an inert gas to increase the pressure
129. Which weak acid has the strongest conjugate base?
(A) acetic acid ($K_a = 1.8 \times 10^{-5}$)
(B) formic acid ($K_a = 1.8 \times 10^{-4}$)
(C) hydrofluoric acid ($K_a = 6.8 \times 10^{-4}$)
(D) propanoic acid ($K_a = 5.5 \times 10^{-5}$)
130. What is the pH of a 0.20 M HA solution ($K_a = 1.0 \times 10^{-6}$) that contains 0.40 M NaA?
(A) 3.15 (B) 3.35 (C) 5.70 (D) 6.30
131. A 0.1 M solution of which salt will have a pH less than 7?
(A) NaCl (B) NH_4Br
(C) KF (D) NaO_2CCH_3
132. What is the $[\text{Mg}^{2+}]$ in 0.10 M NaF that is saturated with MgF_2 at 25 °C?
- | | K_{sp} |
|----------------|----------------------|
| MgF_2 | 6.4×10^{-9} |
- (A) 0.050 M (B) 1.9×10^{-3} M
(C) 1.2×10^{-3} M (D) 6.4×10^{-7} M
133. $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H < 0$
Which change(s) will increase the quantity of $\text{SO}_3(\text{g})$ at equilibrium?
- | | |
|------|--------------------------------------|
| I. | increasing the temperature |
| II. | reducing the volume of the container |
| III. | adding He to increase the pressure |
- (A) I only (B) II only
(C) I and III only (D) II and III only
134. What is the equilibrium expression for the reaction;
 $2\text{ZnS}(\text{s}) + 3\text{O}_2(\text{g}) \rightleftharpoons 2\text{ZnO}(\text{s}) + 2\text{SO}_2(\text{g})$
- (A) $K = \frac{2[\text{SO}_2]}{3[\text{O}_2]}$ (B) $K = \frac{[\text{SO}_2]^2}{[\text{O}_2]^3}$
(C) $K = \frac{2[\text{ZnO}][\text{SO}_2]}{3[\text{ZnS}][\text{O}_2]}$ (D) $K = \frac{[\text{ZnO}]^2[\text{SO}_2]^2}{[\text{ZnS}]^2[\text{O}_2]^3}$
135. What is the pH of a 0.0015 M solution of HNO_3 ?
(A) 1.41 (B) 2.82 (C) 5.65 (D) 11.18

136. In a solution of formic acid ($K_a = 1.7 \times 10^{-4}$), the $[\text{H}^+] = 2.3 \times 10^{-3}$. What is the concentration of formic acid in $\text{mol} \cdot \text{L}^{-1}$?

(A) 7.2×10^{-2} (B) 3.1×10^{-2}
(C) 5.3×10^{-6} (D) 3.9×10^{-7}

137. What is the $[\text{H}^+]$ in a solution in which $[\text{HA}] = 4.0 \times 10^{-2}$ and $[\text{A}^-] = 2.0 \times 10^{-2}$. $[K_a = 3.0 \times 10^{-6}]$

(A) 1.5×10^{-6} (B) 3.0×10^{-6}
(C) 6.0×10^{-6} (D) 3.8×10^{-3}

138. When the compounds below are arranged in order of increasing solubility in water, which order is correct?

	K_{sp}
BaCO_3	2.6×10^{-9}
BaSO_4	1.1×10^{-10}
CaCO_3	4.9×10^{-9}
CaSO_4	7.1×10^{-9}

(A) $\text{BaCO}_3, \text{BaSO}_4, \text{CaCO}_3, \text{CaSO}_4$
(B) $\text{BaSO}_4, \text{CaCO}_3, \text{CaSO}_4, \text{BaCO}_3$
(C) $\text{CaSO}_4, \text{CaCO}_3, \text{BaCO}_3, \text{BaSO}_4$
(D) $\text{BaSO}_4, \text{BaCO}_3, \text{CaCO}_3, \text{CaSO}_4$

139. Which statement is true for a reaction at equilibrium?

(A) All reaction ceases.
(B) The reaction has gone to completion.
(C) The rates of the forward and reverse reactions are equal.
(D) The amount of product equals the amount of reactant.

140. For the hypothetical reaction, $2\text{A}(\text{s}) + \text{B}(\text{g}) \rightleftharpoons 3\text{C}(\text{g})$ what is the equilibrium expression?

(A) $K = \frac{[\text{C}]^3}{[\text{A}]^2[\text{B}]}$ (B) $K = \frac{3[\text{C}]}{2[\text{A}][\text{B}]}$
(C) $K = \frac{[\text{C}]^3}{[\text{A}]^2 + [\text{B}]}$ (D) $K = \frac{[\text{C}]^3}{[\text{B}]}$

141. Acetylsalicylic acid (aspirin) behaves as an acid according to the equation shown. Calculate K_b for the $\text{C}_9\text{H}_7\text{O}_4(\text{aq})$ ion. ($K_a = 3.0 \times 10^{-4}$)

$\text{HC}_9\text{H}_7\text{O}_4(\text{aq}) + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{C}_9\text{H}_7\text{O}_4^-(\text{aq})$
(A) 3.0×10^{-17} (B) 3.3×10^{-11}
(C) 9.0×10^{-8} (D) 3.3×10^3

Equilibrium

142. What will happen to the pH of a buffer solution when a small amount of a strong base is added? The pH will

(A) increase slightly
(B) decrease slightly
(C) remain exactly the same
(D) become 7.0

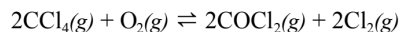
143. When a solution of NH_3 ($K_b = 1.8 \times 10^{-5}$) is titrated with a strong acid the indicator used should change color near a pH of

(A) 1 (B) 5 (C) 9 (D) 13

144. When solid silver chloride (MM = 143.4) is added to 100. mL of H_2O , 1.9×10^{-4} grams dissolves. What is the K_{sp} for silver chloride?

(A) 1.3×10^{-5} (B) 3.7×10^{-6}
(C) 3.7×10^{-8} (D) 1.8×10^{-10}

145. For the reaction,



what is the equilibrium expression, K_c ?

(A) $K_c = \frac{[\text{COCl}_2][\text{Cl}_2]}{[\text{CCl}_4][\text{O}_2]}$ (B) $K_c = \frac{2[\text{COCl}_2][\text{Cl}_2]}{[\text{CCl}_4][\text{O}_2]}$
(C) $K_c = \frac{[\text{COCl}_2][\text{Cl}_2]^2}{[\text{CCl}_4][\text{O}_2]}$ (D) $K_c = \frac{[\text{COCl}_2]^2[\text{Cl}_2]^2}{[\text{CCl}_4]^2[\text{O}_2]}$

146. For the reaction,



Which change(s) will increase the fraction of $\text{SO}_3(g)$ in the equilibrium mixture?

- | |
|-------------------------------|
| 1. Increasing the pressure |
| 2. Increasing the temperature |
| 3. Adding a catalyst |

(A) 1 only (B) 3 only
(C) 1 and 3 only (D) 1, 2 and 3

147. What is the $[\text{H}^+]$ in a 0.10 M solution of ascorbic acid, $\text{C}_6\text{H}_8\text{O}_6$?

K_a	
$\text{C}_6\text{H}_8\text{O}_6$	8.0×10^{-5}

(A) 8.0×10^{-6} M (B) 2.8×10^{-3} M
(C) 4.0×10^{-3} M (D) 5.3×10^{-3} M

148. A 0.10 M solution of which salt is the most acidic?

(A) $\text{NH}_4\text{C}_2\text{H}_3\text{O}_2$ (B) NaCN
(C) KNO_3 (D) AlCl_3

149. A student is asked to prepare a buffer solution with a pH of 4.00. This can be accomplished by using a solution containing which of the following?

K_a	
HNO_2	4.5×10^{-4}
HCN	4.9×10^{-10}

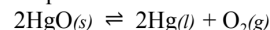
(A) HNO_2 only (B) HCN only
(C) HNO_2 and NaNO_2 (D) HCN and NaCN

150. A saturated solution of which compound has the lowest $[\text{Ca}^{2+}]$?

K_{sp}	
CaF_2	4.0×10^{-11}
CaCO_3	8.7×10^{-9}
Ca(OH)_2	8.0×10^{-6}
CaSO_4	2.4×10^{-5}

(A) CaF_2 (B) CaCO_3 (C) Ca(OH)_2 (D) CaSO_4

151. Mercury(II) oxide, HgO , is decomposed upon heating according to this equation.



What is the equilibrium expression for this process?

(A) $K = \frac{[\text{Hg}]^2[\text{O}_2]}{[\text{HgO}]^2}$ (B) $K = \frac{[\text{Hg}][\text{O}_2]}{[\text{HgO}]}$
(C) $K = [\text{Hg}][\text{O}_2]$ (D) $K = [\text{O}_2]$

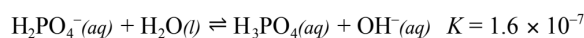
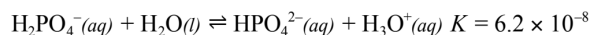
152. Consider this reaction.



What conditions of temperature and pressure will produce the highest yield of NOCl at equilibrium?

	T	P
(A)	high	high
(B)	high	low
(C)	low	high
(D)	low	low

153. The dihydrogen phosphate ion undergoes these reactions in water.



What is the conjugate base of H_2PO_4^- ?

(A) $\text{HPO}_4^{2-}(aq)$ (B) $\text{H}_2\text{O}(l)$
(C) $\text{OH}^-(aq)$ (D) $\text{H}_3\text{PO}_4(aq)$

154. What is the pH of a 0.15 M solution of formic acid, HCOOH ?

Formic Acid	K_a
HCOOH	1.9×10^{-4}

(A) 1.49 (B) 2.27
(C) 3.72 (D) 4.55

Equilibrium

155. Which of these mixtures constitute buffer solutions?

Mixture 1 25 mL of 0.10 M HNO₃ and
25 mL of 0.10 M NaNO₃

Mixture 2 25 mL of 0.10 M HC₂H₃O₂ and
25 mL of 0.10 M NaOH

- (A) 1 only (B) 2 only
(C) both 1 and 2 (D) neither 1 nor 2

156. Which salt gives the most acidic 0.1 M solution in water?

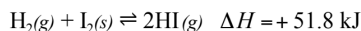
- (A) NaCl (B) NaNO₂
(C) NH₄Cl (D) NH₄NO₂

157. What is the solubility of magnesium carbonate, MgCO₃, in water at 25 °C?

Data for MgCO ₃	
molar mass	84 g·mol ⁻¹
K_{sp} at 25 °C	6.8×10^{-6}

- (A) 0.22 g·L⁻¹ (B) 2.6×10^{-3} g·L⁻¹
(C) 3.1×10^{-5} g·L⁻¹ (D) 8.1×10^{-8} g·L⁻¹

Questions 158 and 159 should both be answered with reference to this system.



158. Which would increase the equilibrium quantity of HI(g)? Assume the system has reached equilibrium with all three components present.

I. increasing pressure II. increasing temperature

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

159. What is the equilibrium constant expression for this system?

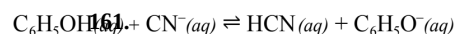
- (A) $K = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$ (B) $K = \frac{[\text{H}_2][\text{I}_2]}{[\text{HI}]^2}$
(C) $K = \frac{2[\text{HI}]}{[\text{H}_2][\text{I}_2]}$ (D) $K = \frac{[\text{HI}]^2}{[\text{H}_2]}$

160. The ionization of benzoic acid is represented by this equation.



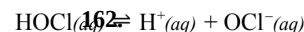
If a 0.045 M solution of benzoic acid has an [H⁺] = 1.7×10^{-3} , what is the K_a of benzoic acid?

- (A) 7.7×10^{-5} (B) 6.4×10^{-5}
(C) 3.8×10^{-2} (D) 8.4×10^{-1}

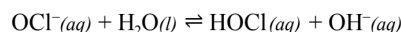


The equilibrium constant for this reaction is less than 1. What is the strongest base in this system?

- (A) C₆H₅OH(aq) (B) CN⁻(aq)
(C) HCN(aq) (D) C₆H₅O⁻(aq)



The ionization of hypochlorous acid represented above has $K = 3.0 \times 10^{-8}$ at 25°C. What is K for this reaction?



- (A) 3.3×10^{-7} (B) 3.0×10^{-8}
(C) 3.0×10^6 (D) 3.3×10^7

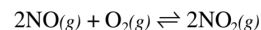
163. Lead(II) fluoride (PbF₂), lead(II) chloride (PbCl₂), lead(II) bromide (PbBr₂), and lead(II) iodide (PbI₂) are all slightly soluble in water. Which lead salt will **increase** in solubility when its saturated solution is acidified?

- (A) PbF₂ (B) PbCl₂
(C) PbBr₂ (D) PbI₂

164. For a system in equilibrium, the rate constant for the forward reaction is represented by k_f and the rate constant for the reverse reaction is represented by k_r . Which equation represents the equilibrium constant for this reaction in the forward direction?

- (A) $K_{eq} = k_f \cdot k_r$ (B) $K_{eq} = \frac{k_f}{k_r}$
(C) $K_{eq} = \frac{k_r}{k_f}$ (D) $K_{eq} = \frac{1}{k_f \cdot k_r}$

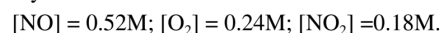
Questions 165 and 166 should both be answered with reference to this reaction, for which ΔH° is negative.



165. Which would increase the partial pressure of NO₂(g) at equilibrium?

- (A) decreasing the volume of the system
(B) adding a noble gas to increase the pressure of the system
(C) removing some NO(g) from the system
(D) adding an appropriate catalyst

166. At a certain temperature the equilibrium concentrations for this system are:



What is the value of K_c at this temperature?

- (A) 0.063 (B) 0.50 (C) 1.4 (D) 2.0

167. What is the pH of a 0.025 M solution of KOH?

- (A) 1.60 (B) 3.69 (C) 10.31 (D) 12.40

Equilibrium

168. What is the $[H^+]$ of a 0.075 M solution of the acid HA?

Equilibrium Constant, K_a	
HA	4.8×10^{-8}

- (A) 6.1×10^{-4} M (B) 2.2×10^{-4} M
(C) 6.0×10^{-5} M (D) 4.8×10^{-8} M

169. Which salt produces the most alkaline solution at a concentration of 0.1 M?

- (A) KNO_3 (B) $MgCl_2$
(C) NH_4Cl (D) $NaNO_2$

170. A 0.052 M solution of benzoic acid, C_6H_5COOH , is titrated with a strong base. What is the $[H^+]$ of the solution one-half way to the equivalence point?

Equilibrium Constant, K_a	
C_6H_5COOH	6.3×10^{-5}

- (A) 6.3×10^{-5} M (B) 1.8×10^{-3} M
(C) 7.9×10^{-3} M (D) 2.6×10^{-2} M

171. A buffer solution made with NH_3 and NH_4Cl has a pH of 10.0. Which procedure(s) could be used to lower the pH?

1. adding HCl
2. adding NH_3
3. adding NH_4Cl

- (A) 1 only (B) 2 only
(C) 1 and 3 only (D) 2 and 3 only

172. How many moles of calcium fluoride, CaF_2 , must be dissolved in 2.0 L of water at 25 °C to form a saturated solution?

K_{sp} at 25 °C	
CaF_2	1.6×10^{-10}

- (A) 2.6×10^{-2} mol (B) 1.3×10^{-3} mol
(C) 6.8×10^{-4} mol (D) 3.4×10^{-4} mol